REQUIREMENT ANALYSIS

Electric Vehicle Market Performance & Insights

Requirements for the Tableau Analytics Dashboard

1. Purpose

This phase defines the data, analytical, and usability requirements for the EV market dashboard. It establishes what the solution should support before dashboard design and development begin.

2. Data Requirements

- Use the provided EV market dataset files as the primary data source.
- Review field names, data types, units, and definitions before building visualizations.
- Identify missing values, duplicate records, inconsistent categories, and invalid values.
- Confirm whether train and test files share the same schema and whether combining them is appropriate for the project objective.
- Use only fields that are present and sufficiently reliable in the supplied data; document any assumptions.

3. Functional Requirements

ID	Requirement
FR-01	Provide a dashboard overview with clearly labelled summary metrics supported by the dataset.
FR-02	Allow users to compare manufacturers using an appropriate market metric.
FR-03	Show category-wise comparisons and sales trends when valid category and time fields are available.
FR-04	Support regional comparisons or a map only when reliable geographic fields are available.
FR-05	Provide views to explore available vehicle attributes such as price, battery capacity, and driving range.
FR-06	Use relevant filters and interactive actions so users can explore subsets of the data.
FR-07	Include a concise Tableau Story or guided sequence to communicate key findings.

4. Non-Functional Requirements

- Clarity: charts, titles, labels, units, and legends should be understandable.
- Consistency: use a consistent visual style, number formatting, and terminology.
- Usability: dashboard navigation and filters should be straightforward.
- Reliability: totals and summaries should match the underlying data and calculation logic.
- Performance: views and filters should respond within a reasonable time for the dataset size.
- Transparency: document important assumptions, missing-data handling, and metric definitions.

5. Intended Users

The dashboard is intended for students, researchers, and users interested in exploring EV market patterns. It may also support preliminary market comparisons by analysts. The dashboard is descriptive and should not be treated as a forecasting or causal-analysis tool unless additional methods and data are introduced.

6. Constraints and Dependencies

- The available dataset determines which metrics and comparisons can be supported.
- Sales trend analysis depends on the availability of a valid time field.
- Geographic analysis depends on usable region or location fields.
- Metric calculations must use documented definitions and consistent units.
- The dashboard is developed using Tableau Public and must work within its supported features and publishing options.

7. Acceptance Criteria

The requirements phase is considered complete when:

- Dataset fields and their meanings have been reviewed.
- The proposed dashboard views can be mapped to available data fields.
- Required filters and interactions have been identified.
- Data quality concerns and assumptions have been recorded.
- The planned dashboard scope is clear enough to proceed to Project Design.

8. Phase Output

The output of this phase is a documented list of data, functional, and non-functional requirements, along with constraints and acceptance criteria. The next phase will translate these requirements into a dashboard layout and visualization design.